

# A Feature-Level Fusion-Based Multimodal Analysis of Recognition and Classification of Awkward Working Postures in Construction

Xiaer Xiahou, Ph.D.<sup>1</sup>; Zirui Li<sup>2</sup>; Jikang Xia<sup>3</sup>; Zhipeng Zhou, A.M.ASCE<sup>4</sup>; and Qiming Li, Ph.D.<sup>5</sup>

**Abstract:** Developing approaches for recognition and classification of awkward working postures is of great significance for proactive management of safety risks and work-related musculoskeletal disorders (WMSDs) in construction. Previous efforts have concentrated on wearable sensors or computer vision-based monitoring. However, certain limitations need to be further investigated. First, wearable sensor-based studies lack reliability due to vulnerability to environmental interferences. Second, conventional computer vision-based recognition demonstrates classification inaccuracy under adverse environmental conditions, such as insufficient illumination and occlusion. To address the above limitations, this study presents an innovative and automated approach for recognizing and classifying awkward working postures. This approach leverages multimodal data collected from various sensors and apparatuses, allowing for a comprehensive analysis of different modalities. A feature-level fusion strategy is employed to train deep learning-based networks, including a multilayer perceptron (MLP), recurrent neural network (RNN), and long short-term memory (LSTM). Among these networks, the LSTM model achieves optimal performance, with an impressive accuracy of 99.6% and an F1-score of 99.7%. A comparison of metrics between single-modality and multimodal-fused training methods demonstrates that the incorporation of multimodal fusion significantly enhances the classification performance. Furthermore, the study examines the performance of the LSTM network under adverse environmental conditions. The accuracy of the model remains consistently above 90% in such conditions, indicating that the model's generalizability is enhanced through the multimodal fusion strategy. In conclusion, this study mainly contributes to the body of knowledge on proactive prevention for safety and health risks in the construction industry by offering an automated approach with excellent adaptability in adverse conditions. Moreover, this innovative attempt integrating diverse data through multimodal fusion may provide inspiration for future studies to achieve advancements.

**DOI:** [10.1061/JCEMD4.COENG-13795](https://doi.org/10.1061/JCEMD4.COENG-13795). © 2023 American Society of Civil Engineers.

**Practical Applications:** Mitigating potential risk factors for work-related musculoskeletal disorders (WMSD) in construction and improving safety and health performance are crucial in construction projects. Construction workers are frequently exposed to prolonged periods of awkward working postures. In pursuit of a more comprehensive solution, a pioneering and automated approach for the recognition and classification of such postures is developed. Specifically, this approach is rooted in the use of joint point data extracted from RGB images, synergistically fused with motion data representing the activity state and electroencephalogram (EEG) data representing the cognitive state. Rigorously tested, this approach demonstrates remarkable classification performance in deep-learning networks, boasting a maximum accuracy of 99.6%. Such high accuracy substantiates its potential for implementation in real construction management. Considering the inherent complexities of dynamic construction sites, compounded by challenging environmental conditions such as insufficient illumination and occlusion, automated identification methods commonly confront limitations in utility. In response, the integrated approach in this study amalgamates the rich information derived from diverse modalities, ensuring a sustained high accuracy rate of 94.9%. This not only demonstrates the exceptional performance of the new approach, but also its generalizability, thereby enabling proactive management of ergonomic and safety risks in construction sites.

**Author keywords:** Awkward working postures; Wearable sensors; Multimodal fusion; Deep learning; Risk management.

<sup>1</sup>Associate Professor, School of Civil Engineering, Southeast Univ., Nanjing 211189, China (corresponding author). Email: [xhcmre@seu.edu.cn](mailto:xhcmre@seu.edu.cn)

<sup>2</sup>Ph.D. Student, School of Civil Engineering, Southeast Univ., Nanjing 211189, China. Email: [seulzr@126.com](mailto:seulzr@126.com)

<sup>3</sup>Undergraduate Student, School of Civil Engineering, Southeast Univ., Nanjing 211189, China. Email: [213202032@seu.edu.cn](mailto:213202032@seu.edu.cn)

<sup>4</sup>Associate Professor, School of Economics and Management, Nanjing Univ. of Aeronautics and Astronautics, Nanjing 210000, China. Email: [zhouzhipeng@nuaa.edu.cn](mailto:zhouzhipeng@nuaa.edu.cn)

<sup>5</sup>Professor, School of Civil Engineering, Southeast Univ., Nanjing 211189, China. Email: [njlqming@163.com](mailto:njlqming@163.com)

Note. This manuscript was submitted on March 21, 2023; approved on August 29, 2023; published online on October 13, 2023. Discussion period open until March 13, 2024; separate discussions must be submitted for individual papers. This paper is part of the *Journal of Construction Engineering and Management*, © ASCE, ISSN 0733-9364.

## Introduction

While construction is a pillar industry that drives economic development for China, it is also considered an accident-prone industry, with a statistically three-fold higher accident rate compared to other industries (Man et al. 2017). As reported by the Ministry of Housing and Urban-Rural Development of China (MHUDC), there were 773 accidents with 904 fatalities in 2019, which represents an increase of 139 accidents and 169 fatalities compared to 2016 (Qi et al. 2022). The nature of construction projects, with their demanding work requirements, poses substantial risks that can directly or indirectly result in numerous injuries and disorders. Overall, the safety situation within the construction industry is not promising, emphasizing the urgent need to optimize management measures.

Among the safety risks that can lead to numerous injuries and disorders, construction workers frequently must maintain prolonged awkward working postures. Such situations commonly refer to positions in which the body deviates significantly from a neutral position (Kittusamy and Buchholz 2004), including bending, climbing, crawling, reaching, and twisting postures (Mora et al. 2016). In the construction industry, these hazardous postures can directly lead to less efficient operation of the muscles or joints, raising the risk of ergonomic injuries and work-related musculoskeletal disorders (WMSDs) (Kong et al. 2018; Antwi-Afari et al. 2022). WMSDs are one of the most common occupational health hazards leading to long-term sickness (Chen et al. 2017) and mainly include diseases of muscles, joints, tendons, and nerves (Nunes and Bush 2012), such as cervical spondylopathy, lumbar and neck muscle strain, and tenosynovitis (ZakerJafari and YektaKooshali 2018; Umer et al. 2018). Furthermore, these WMSDs are contributors to safety accidents, and are attributed to prolonged exposure to intense and repetitive tasks. According to statistics of the Bureau of Labor Statistics (BLS), WMSDs are linked to 32% of all injuries leading to working absenteeism in all industries. In the United Kingdom, WMSDs are responsible for approximately 40% of all work days lost in the construction industry (Antwi-Afari et al. 2018b). In addition, 46% of construction renovation workers are exposed to WMSD risks (Sobeih et al. 2009). Consequently, it is crucial to develop approaches for recognizing and classifying these postures, contributing to effectively preventing factors of WMSD risk in construction and improving safety and health performance.

To develop proactive and effective WMSD risk detection, research on the identification of awkward working postures is continuously conducted from an ergonomic perspective. One method to monitor postures is the manual observation assessment (Antwi-Afari et al. 2022; Kee 2021), which is conducted through the commonly used Ovako Working Analysis System (OWAS) (Brandl et al. 2017), Rapid Entire Body Assessment (REBA) (Hignett and McAtamney 2000), and Rapid Upper Limb Assessment (RULA) (Dockrell et al. 2012) rules. However, these traditional solutions are currently considered to be a high-cost and low-automated approaches. Meanwhile, wearable sensors for measuring motion-related physical metrics are increasingly employed for automated detection combined with computer-based data analysis techniques. Commonly utilized sensor-based techniques include electroencephalogram (EEG) (Jung et al. 2020), electrocardiogram (ECG) (Kumar et al. 2020), and wearable insole pressure system (Kong et al. 2018) methods. With the widespread availability of advanced computer vision-based posture analysis systems in recent years, direct image-based management of ergonomic risks has become feasible (Zhao and Obonyo 2021). Currently, deep-learning-based networks have had excellent performance in automated classification of numerous postures; these methods include convolutional neural network (CNN) (Zhao and Obonyo 2020), multilayer perceptron (MLP) (Antwi-Afari et al. 2018a), recurrent neural network (RNN) (Antwi-Afari et al. 2022), and long short-term memory (LSTM) (Zhao and Obonyo 2020).

However, whether the method is based on wearable sensors or computer vision-based posture monitoring, certain limitations have emerged in existing studies. Wearable sensors can interfere with normal worker operations and have specific requirements for working conditions, resulting in potentially insufficient reliability of recognition results (Antwi-Afari and Li 2018). On the other hand, computer vision-based methods, known for their excellent recognition accuracy and automation, still face challenges in achieving accurate results under adverse environmental conditions such as insufficient illumination and occlusion (Yu et al. 2021). Furthermore,

previous studies on posture identification have primarily focused on single-modality data, which may limit the optimization of results. To overcome these limitations, multimodal fusion technology has been proposed. This technique involves integrating, understanding, and reasoning with multimodal data, which can include various forms of data such as video, audio, and data from different sources or sensors (Majumder et al. 2018). By leveraging the rich information provided by multiple modalities, multimodal fusion aims to enhance the accuracy and robustness of the decision-making process (Zhang et al. 2020). To address the above limitations associated with the complexity of dynamic construction sites (Pan and Zhang 2021), some studies have proposed improved algorithms or approaches to estimate postures under occlusion or other adverse environmental factors (Huo and Hendriks 2012). In contrast, multimodal fusion technology allows for the integrated adoption of these advanced technologies, leveraging their complementary strengths to achieve excellent posture recognition performance (Gandhi et al. 2023). However, the application of this feasible approach in the construction industry is still relatively scarce.

In light of the abovementioned limitations, this study aims to develop an approach for recognizing and classifying awkward working postures using multimodal data collected from wearable sensors and red-green-blue (RGB) cameras. By leveraging feature-level fusion and optimizing deep-learning-based networks, the proposed approach seeks to improve recognition accuracy for specific awkward working postures and enhance performance in adverse environments, such as those with insufficient illumination or occlusion. The developed multimodal-based approach addresses the existing shortcomings to some extent and serves as a reference framework for future studies. Moreover, it can be applied to proactively mitigate WMSD risks in the construction industry. The remainder of this article is organized as follows: section “Background and Related Work” reviews the relevant literature, section “Methodology” provides a detailed description of the experimental settings and data analysis workflow, section “Results and Discussion” presents the data analysis results and corresponding discussions, section “Contributions and Limitations” outlines the implications and limitations, and finally, section “Conclusions” concludes the study.

## Background and Related Work

### *Vision-Based Approaches for Recognizing Work-Related Ergonomic Risks*

To prevent WMSD-related health and safety issues associated with prolonged exposure to intense and repetitive tasks, numerous vision-based technologies have been employed to identify ergonomic risks in construction. Compared to traditional self-reporting methods using questionnaires or interviews (Spielholz et al. 1999), vision-based risk identification can avoid the limitations of high subjectivity and low reliability, regardless of the role in awkward working posture assessment (Spielholz et al. 2001). For vision-based recognition technologies, both manual observation methods and computer-aided vision technologies are commonly considered (Antwi-Afari et al. 2022; Yu et al. 2021).

To implement traditional manual observation methods, systematic observation of worker motion and activity is typically carried out by experienced experts (Dahlberg et al. 2004), where specific data coding and recording rules are utilized (Yu et al. 2021). In addition to the Pose, Activity, Tools, and Handling (PATH) method for systematic observation of activity information (Buchholz et al. 1996), ergonomic risk assessment can be conducted with observational rules and checklists (Vieira and Kumar 2004), including the

Ovaco Working Postures Assessment System (OWAS), the Rapid Upper Limb Assessment (RULA), and the Rapid Entire Body Assessment (REBA). The above ergonomic assessment tools are implemented based on recorded activity and posture data. Although manual observation methods are applicable to the identification of various ergonomic risk factors, they are constrained by substantial cost and time requirements, as well as by the subjectivity and instability of expert judgment (David 2005; Antwi-Afari et al. 2022; Yu et al. 2017).

With the digital upgrading of construction sites, cameras are being deployed more often to monitor the real-time situations of workers and equipment. This hardware base incorporates more computer-aided vision technologies, achieving objective and automated WMSD risk identification (Yu et al. 2021). Specifically, this advanced technology is implemented through different cameras and corresponding computer analysis, including RGB cameras (Yan et al. 2017b), depth cameras (Abobakr et al. 2019), and stereo cameras (Bingöl and Ekinci 2017). In this process, activity information is extracted directly from captured images or videos using artificial intelligence algorithms such as CNN (Zhang et al. 2018), ANN (Antwi-Afari et al. 2018b), and LSTM (Antwi-Afari et al. 2022) networks. Based on ordinary RGB cameras, Yan et al. (2017b) developed a nonintrusive ergonomic posture recognition technique to capture 2D skeleton motions, which aided in construction hazard prevention. Additionally, Wang et al. (2021) proposed a risk assessment approach based on static frame postures collected from a smartphone RGB camera, allowing further evaluation with REBA rules. Moreover, as a markerless motion capture system, Microsoft Kinect allows for extracting a 3D skeleton body from depth video frames (Zhang et al. 2018). Based on this technology, Yu et al. (2017) developed an image-skeleton-based parameterized method, enabling real-time identification of construction worker postures and behaviors. In conclusion, computer-aided vision technologies are relatively low cost and highly automated, enabling accurate ergonomic risk identification. Meanwhile, considering the complexity and dynamics of construction sites (Pan and Zhang 2021), this type of technology is significantly limited by adverse environmental conditions, including insufficient illumination and occlusion (Antwi-Afari et al. 2022; Yu et al. 2021; MassirisFernández et al. 2020).

### Posture Detection with Wearable Sensors

Due to the limitations of noncontact vision-based ergonomic risk identification, not only has computer-aided vision been employed for upgrading, but wearable sensors have also gained popularity for their relatively high applicability on construction sites (Zhao and Obonyo 2020). Through various wearable sensors, critical information related to activity and postures can be precisely collected, including physical data such as acceleration, angular velocity, angle and pressure, as well as physiological data such as EEG, ECG, electrodermal activity (EDA), surface electromyography (sEMG), etc. (Antwi-Afari et al. 2022; Yu et al. 2021). With data from these sensor technologies, it is feasible to identify worker activity in real time, enabling proactive management and intervention for ergonomic risks (Zhang et al. 2018). Moreover, the application of wearable sensors in the process of monitoring, tracking, and recording ergonomic risks can be divided into two main aspects: direct and indirect measurements.

To implement direct measurement of ergonomic risks, existing studies concentrate on the physical data that can directly reflect worker activity and motion. The inertial measurement unit (IMU) sensor, known for its compact size, lightweight design, and real-time data transmission capability via Bluetooth (Antwi-Afari et al. 2022),

stands out as the predominant wearable sensor used in ergonomic risk assessment (Zhang et al. 2018). Utilizing angle data collected from an IMU sensor, Yan et al. (2018) captured workers' trunk inclination angle and holding time, enabling proactive prevention of ergonomic risks. Similarly, Yang et al. (2017) successfully detected and assessed worker gait abnormalities by analyzing gait data collected from an ankle-attached IMU. Moreover, systems of multiple IMU sensors attached to critical body joints are utilized to collect full-body relevant data for analysis (Yu et al. 2021). For instance, Valero et al. (2017) employed a system of eight IMU sensors to measure the angles of critical body joints to identify WMSD-related postures. Although IMUs possess satisfactory accuracy (Schall et al. 2016; Yan et al. 2017a), they require tight attachment to worker bodies during prolonged work to prevent interference from noise, thus imposing a physical burden on normal operation and reducing efficiency (Zhang et al. 2018; Guo et al. 2017). Especially at hot and humid construction sites, the physical burden from tightly attached IMU sensors is increased (Yan et al. 2017b), indicating that additional design and arrangements are urgently needed when using IMUs for the purpose of mitigating such adverse effects on normal worker task performance.

On the other hand, indirect measurements of ergonomic risks in existing studies have collected both physiological and physical data. For the physiological data, there are relatively sufficient studies in construction, with a commitment to explore the association with ergonomic risks. Through a virtual reality (VR) simulation method, Jeon and Cai (2022) conducted EEG-based experiments in an immersive virtual environment, demonstrating the effectiveness of classifying multiple hazards. From another perspective, using measuring spinal biomechanics using sEMG, Antwi-Afari et al. (2017) examined muscle activity and fatigue in various muscle groups during repetitive lifting tasks. Except for physical data such as angle, angular velocity, and acceleration, which can directly reflect activity, other physical data such as insole pressure are also utilized to investigate the association with behaviors, achieving indirect reflection of activity. Antwi-Afari et al. (2022) employed a wearable system named OpenGo to collect insole pressure data and other physical data, achieving accurate recognition and classification of awkward working postures at an actual construction site. As with the aforementioned wearable sensors such as IMUs, the apparatus covered in this section possesses the capability of accurate measurement and similarly interferes with worker normal operation, thus undermining the reliability of the results (Antwi-Afari and Li 2018). Given the high financial cost of the above apparatus, including wearable EEG devices, it is relatively difficult to promote their use for common construction sites.

### Deep-Learning-Based Networks for Prevention of Ergonomic Risks

Automatic and real-time ergonomic risk identification plays a significant role in mitigating WMSDs and safety accidents; managers can access crucial information to facilitate WMSD prevention (Antwi-Afari et al. 2022). As a subfield of artificial intelligence, machine learning is capable of generating models for automatic problem solving based on a given dataset (Baduge et al. 2022). In existing studies on ergonomic risk prevention, machine-learning algorithms have been extensively employed to implement posture classification (Ryu et al. 2022; Zhang et al. 2018), behavior detection (Antwi-Afari et al. 2018a), and psychological monitoring (Jeon and Cai 2022; Jebelli et al. 2018) based on collected data from wearable sensors or cameras. However, traditional machine learning-based classifiers usually demonstrate limitations in higher-dimensional data such as images, video, and

audio (Baduge et al. 2022), as well as large-scale data analysis. Additionally, these classifiers frequently ignore the spatiotemporal context (Antwi-Afari et al. 2022), making them inapplicable for processing time-series data.

Due to its ability to automate tasks and deliver outstanding performance, deep learning has garnered significant attention in the field of construction as a subfield of machine learning. In existing studies related to ergonomic risk prevention in construction, deep learning is mainly concentrated on application scenarios such as object detection (Jeelani et al. 2021; Wu et al. 2019), object tracking (Angah and Chen 2020; Son and Kim 2021), image classification (Yang et al. 2023), and activity recognition (Antwi-Afari et al. 2022; Zhao and Obonyo 2021). To address the limitations in the previous image or video analysis, CNNs have been commonly promoted to identify hazards and behaviors. For instance, Fang et al. (2018) developed two CNN models to detect the presence of workers and identify harnesses, with excellent accuracies of 99% and 80%, respectively. In a subsequent study by Fang et al. (2019), a computer vision-based approach using an R-CNN model was used to detect individuals traversing structural supports, achieving recall and precision rates for overlapping detection of 90% and 75%, respectively. In response to the limitations of processing time-series data, RNNs are widely adopted for processing time-series data, while the RNN variant, LSTM, has received great interest in construction. Using worker motion data collected by IMUs, Yang et al. (2020) developed a bidirectional LSTM (Bi-LSTM) network to realize the classification of physical load levels, with an accuracy of 74.6% to 98.6%. Antwi-Afari et al. (2022) collected relevant data using a wearable insole system and trained RNN-based LSTM, Bi-LSTM, and gated recurrent unit (GRU) networks, with up to 99.01% accuracy in classifying awkward working postures. Moreover, Zhao and Obonyo (2021) achieved excellent performance in recognizing awkward postures by utilizing a CLN model, which integrated a CNN with LSTM layers using data collected from IMUs.

Although ergonomic risk prevention has achieved excellent performance using deep-learning-based networks, most prevailing studies only consider single-modality data, with few studies attempting to consider data from multiple sensor systems simultaneously to improve the reliability of the results (Zhang et al. 2022; Chen et al. 2019). To enhance the recognition and classification of awkward working postures, this work adopts a multimodal fusion strategy to develop networks, along with mitigating the possible adverse effects on workers by designing a multimodal data acquisition system.

## Methodology

The workflow of this study is depicted in Fig. 1. In the first part, Data Acquisition and Processing, details of the adopted apparatus and recruited participants are described, as well as the experimental procedure for data collection. Furthermore, preprocessing and segmentation of the collected data are performed to develop the dataset for the analysis process. Next, the second part, Feature Engineering and Fusion, presents multimodal fusion strategies and multimodal feature engineering. Finally, the third part, Network Training and Performance Evaluation, presents the three selected deep-learning-based network architectures, model training settings and environment, and the utilized metrics for performance evaluation.

### Data Acquisition

#### Participants and Apparatus

A total of nine construction workers were voluntarily recruited to participate in outdoor experiments for data acquisition; they

comprised seven males and two females. The occupations of the participants included bricklayer, concrete worker, rebar worker, and carpenter. In addition, the participants had good physical fitness without any musculoskeletal disease (Antwi-Afari et al. 2022) and had no history of psychiatric disorders or drug intake that could affect brain function. The mentioned basic requirements for participants were proposed to conduct experiments effectively and prevent them from possible musculoskeletal disorders (Wang et al. 2021). Their average age was 37.8 years (SD = 5.51), their average height was 169.6 cm (SD = 4.67), their average weight was 66.0 kg (SD = 8.02), and their average working experience was 12.8 years (SD = 5.91). Before the experiments, all participants were informed of the procedure and completed a demographic information survey (Table 1).

To conduct outdoor experiments for multimodal data acquisition, a dedicated wearable sensor system (Fig. 2) consisting of a wearable EEG device (Emotiv EPOC X), foot pressure sensors, and an IMU sensor was adopted. The widely used Emotiv EPOC X (Emotiv Corporation, San Francisco) was employed to stream the microvoltages of activated brain regions and record the EEG data at a sampling frequency of 128 Hz (Ke et al. 2021; Xing et al. 2020; Jebelli et al. 2018). The EEG device contained 14 channels (AF3, F3, F7, FC5, T7, P7, O1, O2, P8, T8, FC6, F8, F4, and AF4), which were placed on participant scalps according to the international 10–20 system of electrode placement. Foot pressure sensors were utilized to measure the exerted force of participant's feet as a reflection of human activities (mainly focusing on the lower body) (Antwi-Afari et al. 2022). In this study, a total of 12 resistive thin-film pressure sensors were employed to dynamically transfer data via Bluetooth technology at variable frequencies below 10 Hz (device-limited frequency). To measure the acceleration and angular velocity of the  $x$ ,  $y$ , and  $z$  axes, an IMU sensor was placed on the top of the hardhat to work. The motion data were recorded at a frequency of 100 Hz, which is of significance for reflecting activities of the upper body.

Additionally, a smartphone (iPhone13pm) was employed to capture RGB videos in AVI format (at 1,080 p and 30 fps) during experiments to provide data for computer vision-based posture tracking, and a fixed tripod was used to keep the shooting position and angle unchanged (Wang et al. 2021). Furthermore, the each RGB video second was converted to 30 corresponding static pictures with sampling rate of 100%.

### Experimental Procedure and Data Collection

Data acquisition experiments were conducted on the civil engineering experimental platform of Southeast University. With the assistance of the experimenters, the wearable sensor system was installed on the participants and debugged compliantly in preparation for the next stage of implementation. To capture RGB videos containing the whole body image, the smartphone was fixed at 2–2.5 m from the coronal plane of each participant and at a height of 1.5 m above the ground (Wang et al. 2021).

During the experiments, seven participants (five males and two females) were instructed to successively perform the awkward working posture-related operations listed in Table 2, including scaffold installation and reinforcement banding, which frequently appeared in repetitive construction tasks and potentially exposed workers to an elevated risk of WMSDs. Different from previous studies that select these posture-related operations by observing daily repetitive operations of workers (Antwi-Afari et al. 2022; Yu et al. 2021), this study was based on knowledge from site investigations and the literature. Details of awkward working postures and relevant operations are also presented in Table 2.

After the instruction given by the experimenters, the participants began to maintain the specific operation status for 60 s and rested

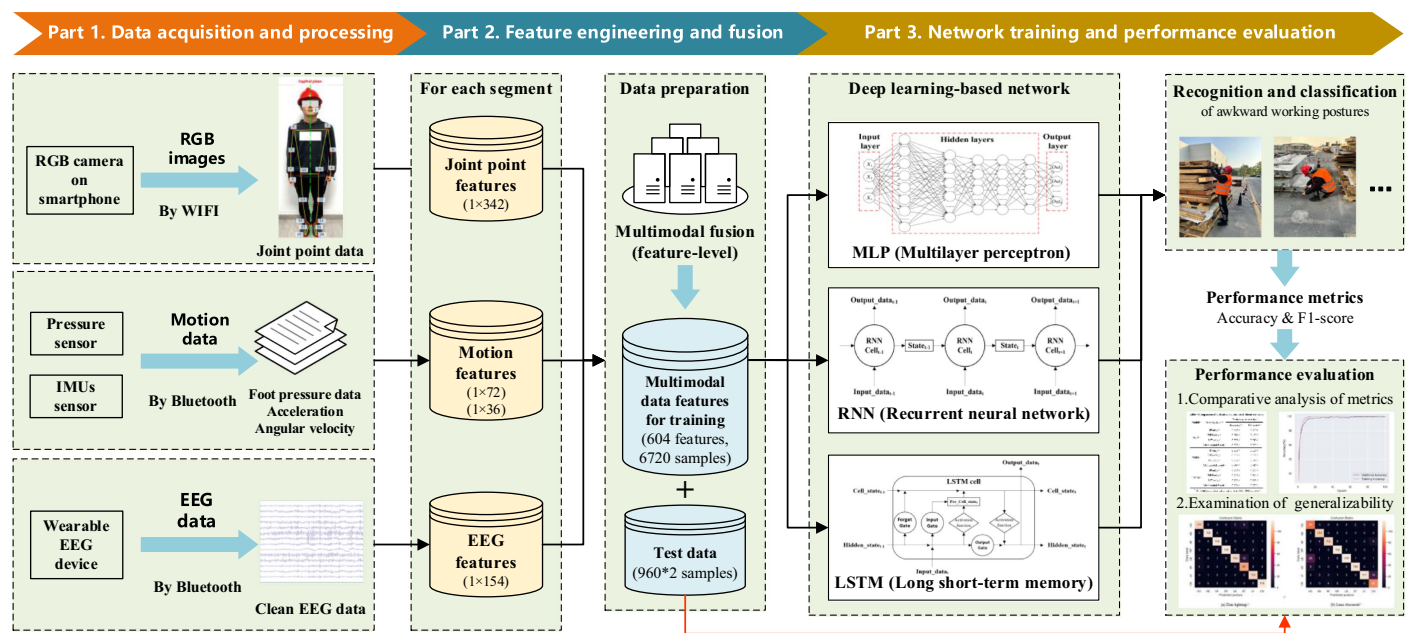


Fig. 1. Overview of the workflow of this study. (Images by authors.)

Table 1. Demographic information of participants

| Statistical parameter | Age (years) | Height (cm) | Weight (kg) | Working experience (years) |
|-----------------------|-------------|-------------|-------------|----------------------------|
| Mean                  | 37.8        | 169.6       | 66.0        | 12.8                       |
| Standard deviation    | 5.51        | 4.67        | 8.02        | 5.91                       |
| Minimum value         | 28          | 163         | 53          | 5                          |
| Maximum value         | 46          | 178         | 76          | 21                         |

for 30 s before the next operation. In this way, the participants completed the assigned experimental tasks in different awkward working postures. Because this study primarily focuses on ergonomic risks associated with awkward working postures during operations, it is assumed that workers are not exposed to any risk during idle time by default. In addition, the wearable sensor system was utilized to record emotion and motion data synchronously, while the RGB camera independently recorded work process videos of each posture. All of the aforementioned experiments were completed under similar experimental conditions, including sunlight illumination, location, surroundings, etc.

To verify the performance of the proposed approach in mitigating the negative effects of adverse environmental conditions, this study additionally conducted two sets of experiments for comparative analysis. Specifically, this study considered the impact of two specific conditions, insufficient illumination and restricted field of view (Yu et al. 2021), on the recognition and classification accuracy. During these experiments, the other two male participants were asked to perform the same procedures described above wearing the wearable sensor system, with the RGB camera maintaining the same position and angle for video capture. In contrast, two additional sets of experiments were conducted under insufficient illumination and occlusion to collect dedicated test data for performance verification. All other experimental settings remained the same, including illumination, location, surroundings, etc., mentioned above.

## Data Preprocessing and Segmentation

After multimodal data acquisition, the next critical stage was data processing, including data preprocessing and segmentation. In this study, the storage data were uniformly transferred onto a laptop for data processing.

### Data Preprocessing

**RGB Images.** Through the smartphone RGB camera, videos were captured and converted into RGB images. Based on the MediaPipe open source framework, which is widely used for posture recognition (Xiao et al. 2023), 19 critical joint points were extracted from each image in static mode. Then, the 3D spatial coordinates ( $x$ -,  $y$ -, and  $z$ -axes) of the landmarks were recorded and standardized to the range of 0–1 as the posture data. Fig. 2 demonstrates the distribution of the 19 joint points extracted using the MediaPipe framework.

**Motion Data.** The motion data were composed of foot pressure data from 12 channels and the acceleration and angular velocity of the  $x$ ,  $y$ , and  $z$  axes. Because the pressure sensor sampling frequency is unstable, anomalous data were checked and eliminated, ensuring controllable errors in posture recognition (Antwi-Afari et al. 2022). In addition, a highpass filter of 6 Hz was used as a cutoff to remove lower-frequency data, ultimately enabling a steady random sampling of six data points per second to form the sequence (Zhao and Obonyo 2020). According to the preprocessing operations applied in existing studies (Zhang et al. 2022), the min-max normalization method was employed to convert the raw pressure data into the range of 0–1 to be more applicable to model training. Given that the acceleration and angular velocity sequences from IMUs were evenly transmitted, no signal artifact removals were specifically implemented (Antwi-Afari et al. 2022). Moreover, no further data normalization was performed considering the spatial properties of these data.

**EEG Data.** Due to the sensitivity and randomness of EEG data, extrinsic and intrinsic artifacts frequently exert a negative influence on data quality (Jeon and Cai 2021, 2022). In this study, a commonly used toolbox named EEGLAB for EEG signal preprocessing was utilized. Considering the Nyquist frequency and frequency



**Fig. 2.** Distribution of the 19 critical joint points extracted using the MediaPipe framework. (Images by authors.)

**Table 2.** Details of the selected awkward working posture-related operations

| Number | Operation                        | Relevant posture                          | Posture label | Occupational risk factor                       | Hidden risks                                    | Identifying sources                                    |
|--------|----------------------------------|-------------------------------------------|---------------|------------------------------------------------|-------------------------------------------------|--------------------------------------------------------|
| 1      | Scaffold installation            | Hands overhead                            | HO            | Prolonged extending arms above shoulder height | Thoracic outlet syndrome                        | Ying et al. (2021) and Cutlip et al. (2000)            |
| 2      | Rebar tying                      | Knees bending                             | KB            | Sustained bending of knees                     | Osteoarthritis and cartilage injury             | Umer et al. (2017, 2018) and Antwi-Afari et al. (2022) |
| 3      | Soldering                        | Neck bending forward                      | NF            | Sustained bending of neck                      | Cervical spondylopathy and neck muscle strain   | Toyoshima et al. (2023)                                |
| 4      | Bricklaying                      | Waist bending forward                     | WF            | Sustained bending of waist forward             | Lumbar muscle strain and lumbar disc herniation | Roberts et al. (2020) and Ryu et al. (2021)            |
| 5      | Beam side plastering             | Leaning backward                          | LB            | Sustained bending of waist backward            | Lumbar muscle strain and lumbar disc herniation | Roberts et al. (2020)                                  |
| 6      | Ceiling plastering and unloading | Body twisting                             | BT            | Prolonged twisting                             | Chronic lumbar muscle strain                    | Roberts et al. (2020) and de Vries et al. (2021)       |
| 7      | Cleaning and housekeeping        | Leaning forward with single-leg supported | LF            | Sustained hyper extension of arms              | Tendonitis, hip arthritis, and tenosynovitis    | Zhao and Obonyo (2021)                                 |
| 8      | Conduit installation             | Curled up in narrow area                  | CN            | Prolonged restricted posture                   | Tension neck syndrome                           | From site investigation                                |

**Table 3.** Overview of the processed multimodal dataset

| Experiment                  | Participants         | Experimental conditions                      | Samples | Data type                                               | Data size<br>(for each sample) |
|-----------------------------|----------------------|----------------------------------------------|---------|---------------------------------------------------------|--------------------------------|
| Data acquisition experiment | 5 males<br>2 females | Sufficient illumination<br>without obscuring | 6,664   | Joint point data                                        | $30 \times 57$                 |
|                             |                      |                                              |         | EEG data                                                | $128 \times 14$                |
|                             |                      |                                              |         | Pressure data (motion data)                             | $6 \times 12$                  |
|                             |                      |                                              |         | Acceleration and angular velocity<br>data (motion data) | $100 \times 6$                 |
| Validation experiments      | 1 male               | Insufficient illumination                    | 952     | Joint point data                                        | $30 \times 57$                 |
|                             |                      |                                              |         | EEG data                                                | $128 \times 14$                |
|                             |                      |                                              |         | Pressure data (motion data)                             | $6 \times 12$                  |
|                             |                      |                                              |         | Acceleration and angular velocity<br>data (motion data) | $100 \times 6$                 |
|                             | 1 male               | Occlusion                                    | 952     | Joint point data                                        | $30 \times 57$                 |
|                             |                      |                                              |         | EEG data                                                | $128 \times 14$                |
|                             |                      |                                              |         | Pressure data (motion data)                             | $6 \times 12$                  |
|                             |                      |                                              |         | Acceleration and angular velocity<br>data (motion data) | $100 \times 6$                 |

range of brain waves, the higher and lower cutoff frequencies of the finite impulse response (FIR) bandpass filter were set to 50 Hz and 1 Hz, respectively, and a notch filter targeting 50/60 Hz was conjunctively adopted to remove the extrinsic artifacts. In addition, through the runcia algorithm and ADJUST plugin, independent component analysis (ICA) was implemented to identify and remove intrinsic artifacts. Based on the above operations, the pure EEG data were exported and prepared for further extraction of features.

### Data Segmentation

Because the sampling frequencies in this study were different, the size of the multimodal data (including joint point coordinates, EEG data, and motion data) obtained in the same 60-s experiments presented significant differences (Table 3). Therefore, as the foundation of deep-learning network training, data segmentation was implemented to obtain segments from the continuous stream of collected data, which effectively indicates the activities and postures of participants within the same time period (Antwi-Afari et al. 2022). During data segmentation, a sliding window technique was utilized to divide different sizes of collected data into segments, and the segments were further labeled in batches manually with reference to the corresponding awkward working postures in the experiments. In this study, the dataset for two stages of experiments contained three types of data corresponding to eight specific operations. According to relevant studies on posture recognition (Zhao and Obonyo 2020), a window size between 0.5 and 2.5 s is feasible for human activity recognition, and can achieve excellent performance (Antwi-Afari et al. 2018a). Therefore, a window size of 1 s was adopted for data segmentation in this study, and 50% window overlap was utilized to avoid transitional posture data loss from adjacent windows and reduce the potential fault caused by transition noise (Zhao and Obonyo 2020).

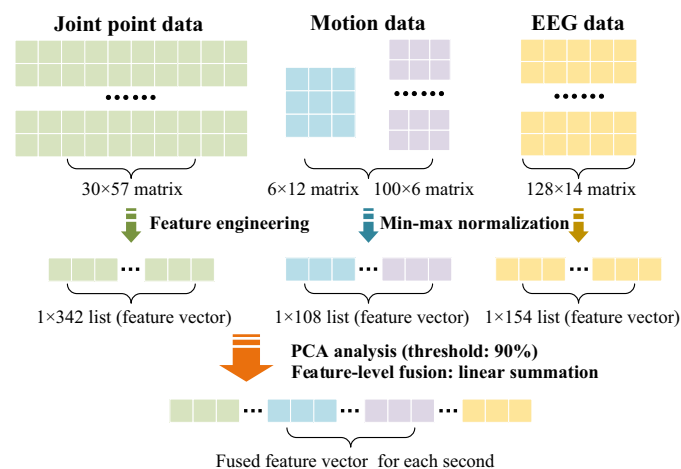
### Dataset

Through preprocessing and data segmentation using a 1-s sliding window with 50% overlap, the multimodal data of each participant were divided into 119 segments. Because there were seven and two participants in the data acquisition and validation experiments, respectively, the total numbers of training data and test data samples for this study were 6,664 (119 segments of seven participants under eight types of postures) and 1,904, respectively, considering the eight awkward working posture-related operations. As there were seven participants in the data acquisition experiment and two

participants in the two validation experiments, the total numbers of training data and test data samples for this study were 720 and 480, respectively. In one data segment, there were three types of multimodal data, including a  $30 \times 57$  matrix of joint point data (30 RGB images/s, 19 extracted joint points, and three spatial axes), a  $128 \times 14$  matrix of EEG data (sampling frequency of 128 Hz and 14 channels), a  $6 \times 12$  matrix of pressure data (randomly sampling six data points/s and 12 sensors), and a  $100 \times 6$  matrix of acceleration and angular velocity data (sampling frequency of 100 Hz and three spatial axes).

### Multimodal Fusion

According to the multimodal dataset depicted in Table 3, the joint point data, EEG data, and motion data differed significantly in data size, such that they could not be directly input to the neural network for training. To address this, a workflow (Fig. 3) was developed, consisting of two critical sections: multimodal fusion strategies and multimodal feature engineering. The multimodal fusion strategy is the critical path to fuse the different sizes of multimodal data, while multimodal feature engineering is the critical foundation for implementing the multimodal fusion strategy, which likewise contributes to transforming the data into a facilitated form for network training (Boeschoten et al. 2023).

**Fig. 3.** Workflow of feature-level multimodal fusion.

## Multimodal Feature Engineering

In accordance with the requirements of multimodal fusion for feature format and recognizability, feature engineering was implemented to lay the foundation, including filtering, engineering, and combining required features (Gandhi et al. 2023). This process extracts useful features from data segments through knowledge or transformation methods, which contributes to model training and performance improvement (Boeschoten et al. 2023). On this basis of data segmentation, the details of feature extraction are conducted automatically using Python 3.9.12, and are described below.

**Joint Point Features.** In each segment, the joint point data were a  $30 \times 57$  matrix, including the  $x$ -,  $y$ -, and  $z$ -axis spatial coordinates of 19 major joints at 30/s. In the feature extraction process, features were calculated based on a sequence of 30 data points from a certain channel (e.g., the  $x$ -coordinate of the right knee). In this study, six types of time-domain features were considered (Antwi-Afari et al. 2018a), including the average, median, standard deviation, variance, skewness, and kurtosis. After feature extraction, the  $30 \times 57$  matrix was converted into a  $6 \times 57$  matrix; meanwhile, the six features of each channel were juxtaposed instead of being arranged in linear temporal logic. Therefore, the matrix for joint point features of one segment was further transformed into a  $1 \times 342$  list (i.e., one row of data) with a corresponding awkward working posture label from experiments, which was input for network training.

**Motion Features.** In each segment, the motion data consisted of a  $6 \times 12$  matrix of pressure data and a  $100 \times 6$  matrix of acceleration and angular velocity data, including sequences of data from the corresponding individual channels. Considering the absence of chaos and fractal properties in motion data (Roy et al. 2020), this study extracted the average, median, standard deviation, variance, skewness, and kurtosis of each channel. In addition, due to the insufficient pressure data in each segment, the statistical indicators, skewness and kurtosis, were excluded for feature extraction. For the alternative, the maximum and minimum values of the data segment were calculated. After feature extraction for the motion data, the  $6 \times 12$  matrix of pressure data and the  $100 \times 6$  matrix were transformed into a  $1 \times 72$  list and  $1 \times 36$  list, respectively. Then, all feature matrixes were converted to forms directly adaptive to the implementation of feature-level fusion.

**EEG Features.** In each segment, the EEG data were a  $128 \times 14$  matrix consisting of 128 data points from each of the 14 channels per second. Considering that the unstructured information of EEG data cannot be directly utilized for further network training, it is necessary to transform preprocessed EEG data into structured features, which can effectively reflect the informative and measurable nature (Jebelli et al. 2018). In view of the above considerations, this study extracted commonly used time domain (i.e., average, standard deviation, and kurtosis), frequency domain (i.e., relative power of five brain waves), and nonlinear domain (i.e., approximate entropy, fuzzy entropy, and Hurst exponent) features to represent each participant's cognitive state. After extracting 11 types of features from 14 channels (i.e., 154 EEG features), the  $128 \times 14$  matrix was directly converted into a  $1 \times 154$  list with a corresponding posture label.

## Multimodal Fusion Strategy

Multimodal fusion, which refers to the integration of multimodal data, enables the incorporation of abundant information and enhances the performance of the model (Zhang et al. 2020). Corresponding fusion strategies can be broadly classified into multiple categories (Gandhi et al. 2023), including early fusion (feature-level fusion), late fusion (decision-level fusion), hybrid fusion, and others.

Given the availability of off-the-shelf feature extraction techniques and the potential correlation between different modalities,

the multimodal fusion strategy employed in this study was early fusion, also known as feature-level fusion. Early fusion focuses on exploring the association between multimodal data (Poria et al. 2017) and converting extracted features from different modalities into a single feature vector (Gandhi et al. 2023). In addition, this feature-level fusion strategy allows early correlation between multi-dimensional features to achieve excellent performance (Majumder et al. 2018). To achieve this fusion process, an applicable linear summation method was utilized and the weights of all modalities were set to be the same, considering the absence of prior knowledge for fusion. Before the implementation, the data of each modality were processed with min-max normalization to effectively reduce data skewing. Subsequently, principal component analysis (PCA) was used to eliminate redundant features under different modalities, aiming to avoid overfitting problems caused by excessive dimensionality (Thakur and Biswas 2022). Moreover, the filtering of features was controlled by the cumulative variance contribution rate, with the threshold set at 90%. This ensured that the retained features accounted for at least 90% of the total variance in the data, allowing for effective dimensionality reduction while preserving a significant amount of information. After filtering by PCA, the 3D modalities retained a total of 128 joint point features, 64 motion features, and 40 EEG features, respectively. The detailed implementation process of feature-level fusion is illustrated in Fig. 3.

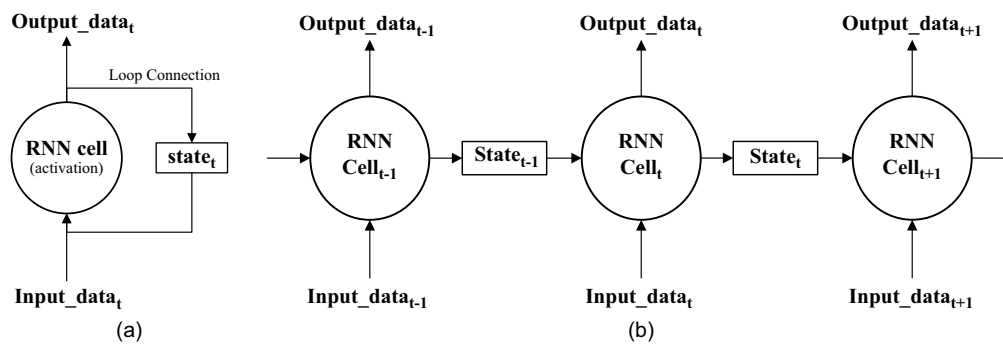
## Development of Deep-Learning-Based Network

### Deep-Learning-Based Network Model Architectures

To classify different types of awkward working postures, deep-learning-based networks were trained to learn unique signal patterns from multimodal data based on extracted features. Taking into account that the input-fused feature data had already undergone one round of preprocessing, it was not certain that excellent performance could be achieved, even with a simple network. With reference to network selection mechanism in previous studies (Antwi-Afari et al. 2018a, 2022), three networks with different complexity levels were ultimately selected (from low to high), including MLP, RNN, and LSTM. Moreover, the input-fused feature data had already undergone one round of preprocessing, leading to potential uncertainty of excellent performance with even a simple network.

**MLP.** MLP is a robust feed-forward artificial neural network (ANN) method, composed of multiple fully connected node layers (including an input layer, hidden layers, and an output layer). In this study, the input layer of trained MLP consists of the extracted features of multimodal data at the specific window size. The number of hidden layers was set to 5, including 600, 350, 120, 90, and 32 neurons, respectively. Additionally, the number of neurons in the output layer was set to match the selected eight types of postures. With respect to the selection of activation functions, the commonly used ReLu function was implemented for each hidden layer, while the softmax function, which is applicable to multiclass classifications, was implemented for the output layer. Considering possible overfitting, regularization was conducted to decrease the complexity of the trained model. To optimize the network, the PyTorch-based CrossEntropyLoss and Adam were used as the loss function and the optimizer, respectively. Moreover, the batch size and epoch of the MLP were set to 32 and 50 by default.

**RNN.** RNN is particularly suitable for datasets that contain sequential information (Sarkar et al. 2022). Its critical working principle involves incorporating previous output into the input layer of the current step, enabling to make predictions based on the context and temporal dependencies of the data. In this study, the RNN training was performed by traversing all data segments, while the basic



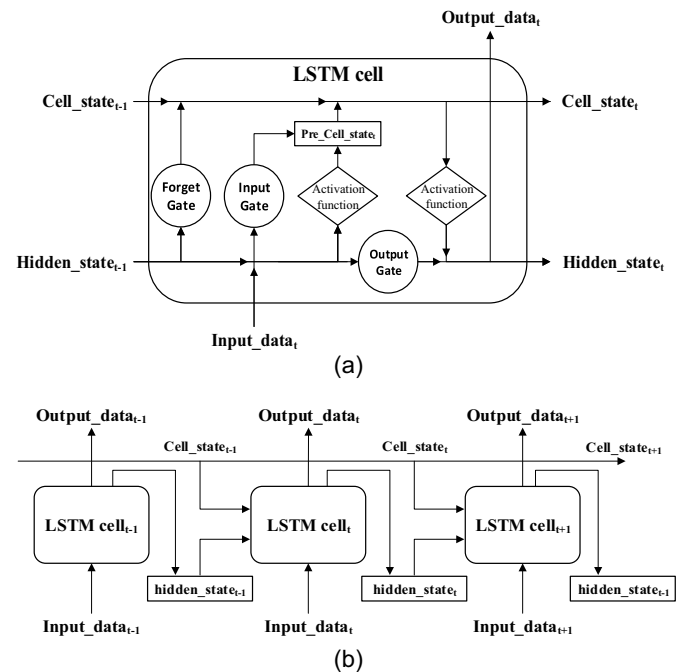
**Fig. 4.** Overview of the RNN architecture: (a) architecture of an RNN cell; and (b) architecture of an unfolded RNN.

architecture of each stage included an input, output, activation function, and a recurrent loop. Figs. 4(a and b) demonstrate the architecture of an RNN cell and an unfolded RNN over time. Similar to the MLP settings, the ReLU activation function was applied to the hidden layers of the RNN. Moreover, regularization was implemented to prevent overfitting, and the PyTorch-based CrossEntropyLoss loss function was utilized. In contrast, the PyTorch-based SGD optimizer was selected for RNN training optimization. In terms of architecture, the RNN consisted of an input layer, two hidden layers (including 100 neurons), and an output layer (using the softmax activation function). Moreover, the batch size and epoch of the RNN were set to 32 and 10 by default.

**LSTM.** LSTM is an upgrade of RNN that incorporates memory cells that can handle the vanishing and exploding gradients commonly encountered in basic RNN. Based on an added gating mechanism, this network can solve temporal sequences and long-term dependency problems (Antwi-Afari et al. 2022). Compared with RNN, LSTM adds the memory cells across multiple stages, which can store, update, and transfer previous outputs to the current stage. In this study, the LSTM cell for dealing with each data segment consisted of an input gate, forget gate, output gate, and cell state. The three gates were employed to determine the new information stored in the memory cells, the forgotten information, and the output information from the output gate. The cell states were critical for transmitting information between adjacent LSTM cells, whose modification process was implemented by the three gates. Figs. 5(a and b) represent the architecture of an LSTM cell and an unfolded LSTM. With reference to RNN settings, the softmax activation function was adopted for the output layer, regularization was implemented to prevent overfitting, and the PyTorch-based Cross-EntropyLoss loss function was utilized. For LSTM training optimization, the PyTorch-based Adam optimizer was used. Moreover, all of the hidden layers of LSTM employed the tanh activation function (with 100 neurons), and the batch size and epoch of LSTM were set to 128 and 20 by default.

### Deep-Learning Model Training

In this study, a multimodal analysis-based approach for awkward working posture recognition and classification was developed, aiming to improve the accuracy of posture identification and applicability in adverse environments. Therefore, the dataset from the data acquisition experiment was designated the “train” dataset, while the dataset from the two validation experiments (insufficient illumination and occlusion) was designated the “test” dataset. Based on the above settings, the neural networks for awkward working posture classification were trained, and the performances under different adverse environments were tested to verify the effectiveness and superiority of the proposed approach. Moreover,



**Fig. 5.** Overview of the LSTM network architecture: (a) architecture of an LSTM cell; and (b) architecture of an unfolded LSTM.

the “train” dataset was further split into “training” and “validation” datasets using a commonly used ratio of 8:2 (Zhao and Obonyo 2021). The specified splitting ratios reserved sufficient data for training the networks.

To implement the algorithm for data processing and model training in this study, a workstation was utilized with an Intel(R) Xeon (R) Silver 4112 2.60 GHz CPU, 64 GB RAM, GPU Nvidia Quadro P4000 (Pascal framework), and Windows 10 OS. In addition, the selected three deep-learning models were implemented using the PyTorch framework (1.12.1 version) and CUDA compute platform (11.3 version) based on the Python (3.9.12 version). Feature engineering was conducted in PyCharm IDE (2022.2.2 version).

### Model Performance Evaluation

The final step involved evaluating the performance of the developed models using multimodal data. Specifically, this study calculated and compared the relevant performance metrics of the trained MLP, RNN, and LSTM network to determine the best model and parameters for classifying posture. The learning rate of the optimizers, which is considered a crucial hyperparameter to fine-tune

for model training (Smith 2017), was set to 0.0001 by default and continuously adjusted to find the optimal state. Notably, the trained models were evaluated via 10-fold cross validation to test the classification performance and select the optimal hyperparameters (Antwi-Afari et al. 2022).

To validate the effectiveness of the proposed multimodal analysis approach, a comparative analysis based on the output performance metrics under multimodal-fused data and single-modality data was conducted. Additionally, two commonly used performance metrics were adopted in this study, the accuracy [Eq. (1)] and averaged F1-score [Eq. (4)]. The accuracy is the percentage of correctly predicted observations regarding the total number of observations and is the most significant metric of classification performance. The F1-score is defined as the harmonic mean of the precision [Eq. (2)] and recall [Eq. (3)], while the averaged F1-score calculated by averaging the F1 scores for all categories is commonly used for performance evaluation of multicategory posture recognition (Zhao and Obonyo 2020, 2021)

$$\text{Accuracy} = \frac{\sum_{i=1}^8 \text{TP}_i}{\sum_{i=1}^8 \text{TP}_i + \text{FP}_i} \quad (1)$$

$$\text{Precision}(i) = \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i} \quad (2)$$

$$\text{Recall}(i) = \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i} \quad (3)$$

$$(\text{averaged}) \text{ F1 score} = \frac{1}{N} \times \sum_i \frac{2 * \text{Precision}(i) \times \text{Recall}(i)}{\text{Precision}(i) + \text{Recall}(i)} \quad (4)$$

where  $\text{TP}_i$  (true positive) is the total number of correctly predicted observations for class  $i$ ;  $\text{FP}_i$  (false positive) is the total number of incorrectly predicted observations for class  $i$ ; and  $\text{FN}_i$  (false negative) is the total number of incorrectly rejected observations for class  $i$ . In addition,  $i$  represents the class label from 1 to 8, while  $N$  [Eq. (4)] denotes the total number of label classes.

## Results and Discussion

The primary objective of this study was to optimize the recognition performance of deep learning-based models for awkward working posture, which was achieved by fusion of multimodal data. In this section, the performance results of the trained MLP, RNN, and LSTM models are presented. By analyzing the output performance of different models using different combinations of training data, a comparative analysis was to assess the effectiveness of the proposed multimodal-based approach. In addition, the performance

was further evaluated under influence of adverse environmental conditions to examine the model generalization.

### Comparative Analysis of Performance Metrics

#### 1. Validation of the Effectiveness of Multimodal Fusion

Detailed results on the performance are presented in Table 4. The accuracy and F1-score of the MLP, RNN, and LSTM models are presented under seven combinations of training data, including single modality of data (including three types), combination of pairwise modalities (including three types), and multimodal-fused data (i.e., fusion of the three-modality data). Notably, the input data of all sets had undergone normalization and PCA analysis (with threshold set at 90%), and the combination of pairwise modalities was completely in accordance with the utilized fusion strategy and corresponding settings. The size of input layer for each combination of training data is also listed in Table 4, corresponding to the size of feature vector after PCA analysis. Moreover, the trained models were evaluated via 10-fold cross validation to test and select optimal hyperparameters (Antwi-Afari et al. 2022).

Obviously, the three models under multimodal-fused data yielded performance greater than 60%, with MLP, RNN, and LSTM accuracies of 87.5%, 91.4%, and 99.6%, respectively, while corresponding F1-scores were 87.5%, 91.5%, and 99.7%. Overall, the results in Table 4 demonstrate that the performance of models under multimodal-fused data was significantly superior to those under other combinations of training data. This directly demonstrated that model training through multimodal fusion yielded relatively high classification performance, further validating the effectiveness of the proposed methodology in the section “Methodology.” In contrast, the models trained with a single modality of data exhibited a consistent pattern in performance: The models trained solely with joint point data demonstrated the highest performance, while those trained exclusively with EEG data exhibited the lowest performance. This difference in performance was attributed to the point coordinates extracted from the pictures being directly related to posture, whereas EEG data, as the measurement of emotional and cognitive state, was only able to indirectly reflect postures.

#### 2. Discussion of Intermodality Incompatibility

Although the fusion strategy was generally expected to improve performance, it is possible that intermodality incompatibility hinders its effectiveness. Specifically, the performance of models trained with the combined data of “EEG + MT” or “EEG + JP” was worse than any of the models trained with single-modality data. This pattern may emerge because EEG data, while indirectly associated with behavioral postures, may not be strongly correlated or complementary to the activity and behavior. As a result, the fusion process did not effectively

**Table 4.** Comparison of performance metrics for the different models

| Training data    | Input layer size | MLP_performance |          | RNN_performance |          | LSTM_performance |          |
|------------------|------------------|-----------------|----------|-----------------|----------|------------------|----------|
|                  |                  | Accuracy        | F1-score | Accuracy        | F1-score | Accuracy         | F1-score |
| JP only          | 128              | 0.773           | 0.700    | 0.806           | 0.805    | 0.926            | 0.925    |
| EEG only         | 64               | 0.519           | 0.457    | 0.882           | 0.825    | 0.909            | 0.894    |
| MT only          | 40               | 0.589           | 0.518    | 0.862           | 0.842    | 0.956            | 0.963    |
| JP + EEG         | 192              | 0.778           | 0.762    | 0.852           | 0.852    | 0.936            | 0.936    |
| JP + MT          | 168              | 0.788           | 0.780    | 0.901           | 0.892    | 0.937            | 0.937    |
| EEG + MT         | 104              | 0.506           | 0.506    | 0.883           | 0.883    | 0.899            | 0.899    |
| Multimodal-fused | 232              | 0.875           | 0.875    | 0.914           | 0.915    | 0.996            | 0.997    |

Note: JP = joint point data; EEG = EEG data; and MT = motion data.

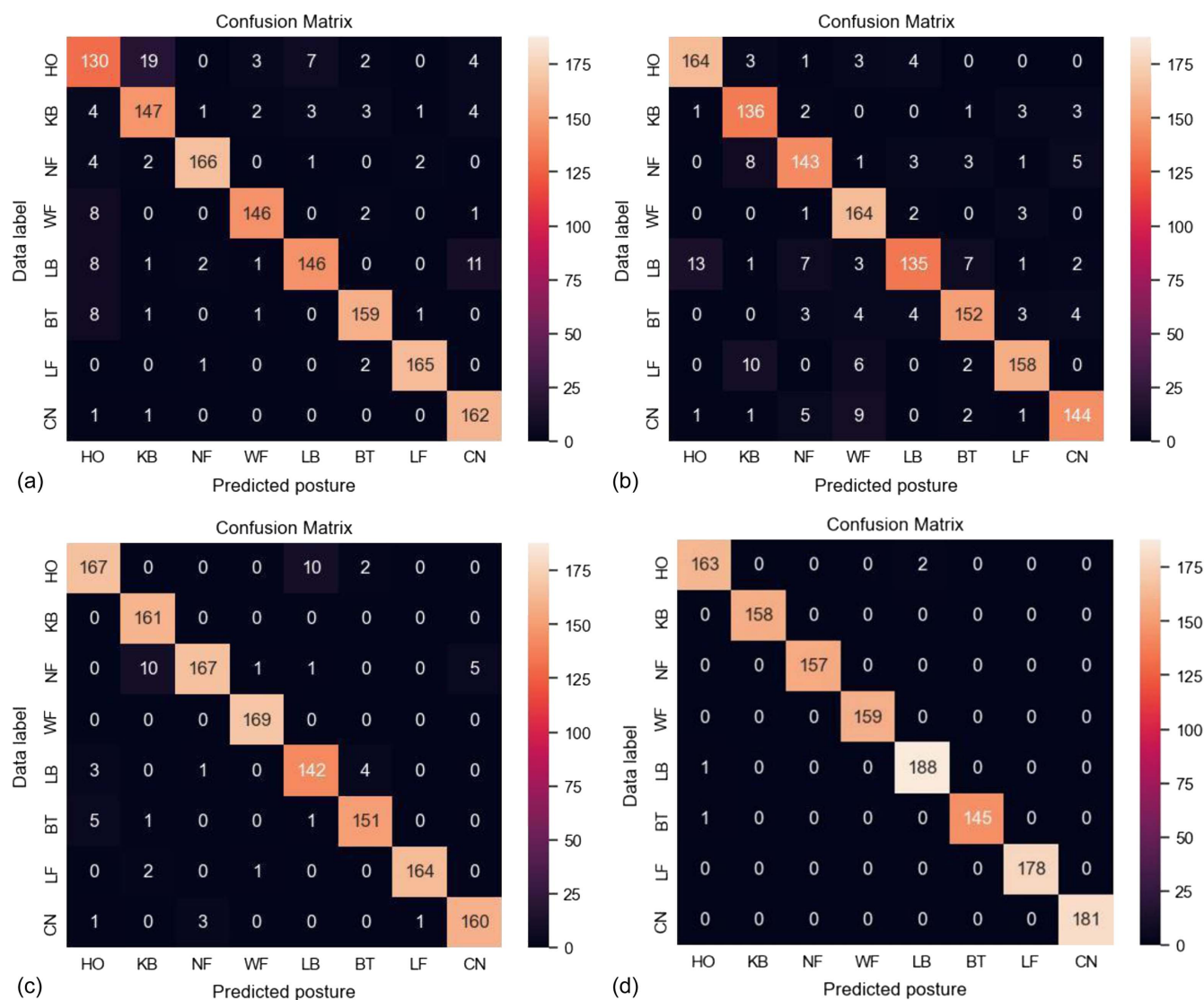
leverage the information from both modalities, leading to a lack of efficiency and improvement in model performance. In conclusion, the results suggested that the EEG data may be redundant information for awkward working postures, due to its low correlation and complementation. However, optimal performance was still achieved by integrating information of the three-modalities data, as it partially mitigated adverse effects of this redundant data modal.

### 3. Further Analysis of LSTM with Optimal Performance

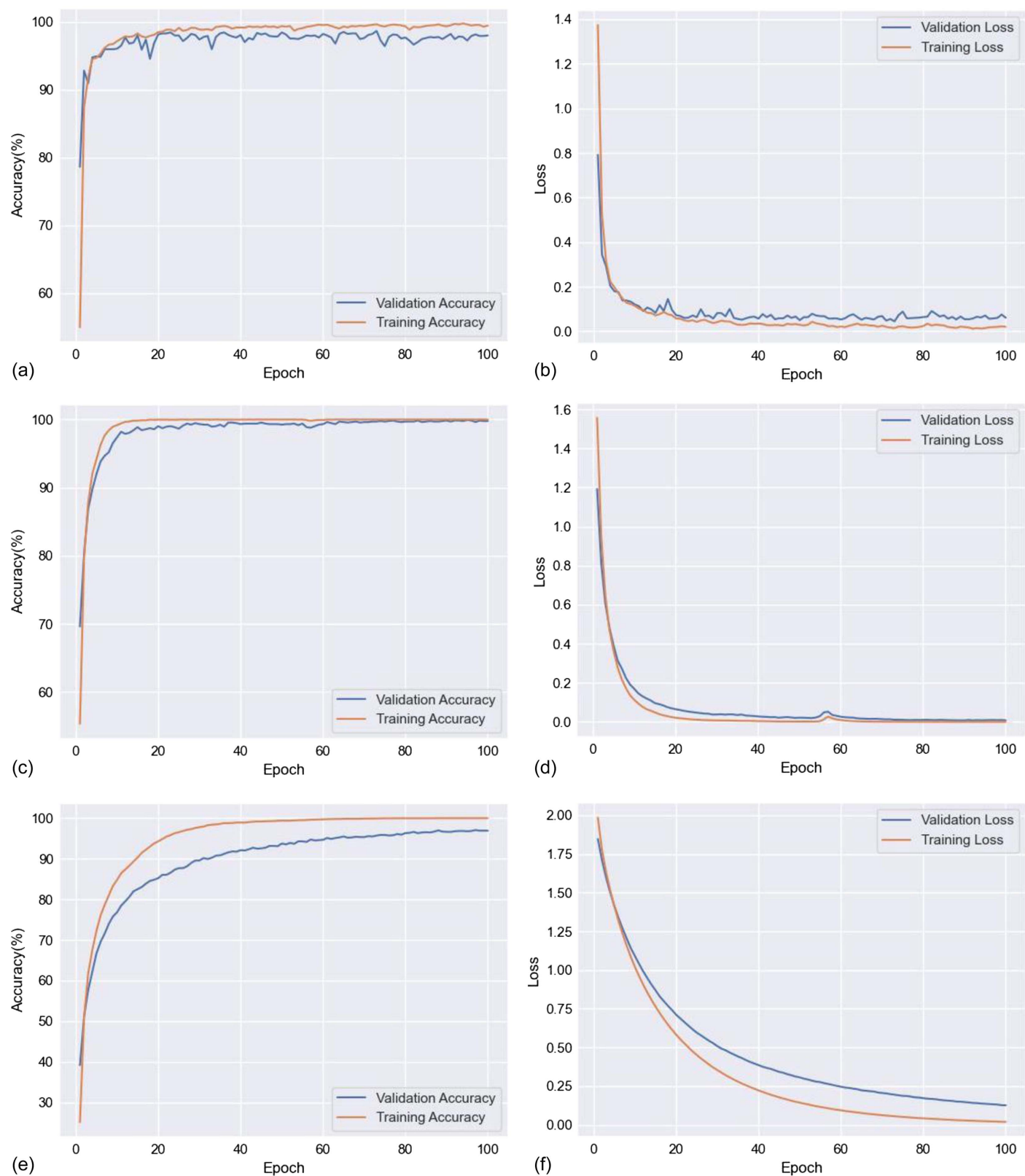
In addition, the results demonstrated that the LSTM model generally achieved optimal performance compared with all trained models. When training with multimodal-fused data to classify different types of postures, the LSTM model achieved an accuracy of 99.6% and an F1-score of 99.7%. To further analyze the misclassifications or confusion between postures, confusion matrices were established for the LSTM model using different modalities of training data. Considering the purpose of highlighting multimodal fusion rather than pairwise combinations, Fig. 6 illustrated these matrixes under single-modality data and multimodal-fused data. In these matrixes, the correct

classes on the diagonals were highlighted to directly reflect the prediction performance for each posture. Under different modalities of training data, the LSTM models differ in the precision of their posture prediction. For instance, using only joint point data, there were 33 incorrectly predicted observations and 35 incorrectly rejected observations for the prediction of the HO (hands overhead) posture. In contrast, using only EEG data, there were 14 incorrectly predicted observations and 11 incorrectly rejected observations for the HO posture. When training with multimodal-fused data, the prediction performance for different postures was adequately optimized, resulting in minimal incorrect predictions or rejections. As shown in Fig. 6(d), only four samples were incorrectly predicted as HO and LB (leaning backward) postures.

The trained models were further evaluated using 10-fold cross validation to test and select hyperparameters. Fig. 7 shows the accuracies and losses over iteration curves of the LSTM model with multimodal-fused data. In general, the LSTM model performance trend was toward increasing accuracy, decreasing loss, and eventually convergence in both training and validation.



**Fig. 6.** Confusion matrixes of the LSTM model using different training data modalities: (a) joint points data only; (b) EEG data only; (c) motion data only; and (d) multimodal-fused data.



**Fig. 7.** Accuracies and losses over iteration curves of the LSTM model under multimodal-fused data: (a) accuracies when the learning rate is 0.001; (b) losses when the learning rate is 0.001; (c) accuracies when the learning rate is 0.0001; (d) losses when the learning rate is 0.0001; (e) accuracies when the learning rate is 0.00001; and (f) losses when the learning rate is 0.00001.

Additionally, a comparison of the final converged values (accuracies and losses) at different learning rates revealed that the optimal performance was obtained when the learning rate was 0.0001 (default setting), satisfying minimal training loss and optimal

validation accuracy. Furthermore, the curves indicated that the accuracies and losses ultimately converged at the 60th epoch when the learning rate was 0.0001, indicating that the LSTM model was effectively trained without overfitting multimodal-fused data.

## Examination of Model Generalizability in Adverse Environmental Conditions

To further assess the generalizability of the LSTM model in adverse environmental conditions, the multimodal data from the two validation experiments were utilized to evaluate its performance. The prediction accuracy metric was selected as the measurement criterion to assess the model output. The four combinations of test data, including multimodal-fused data and single-modality data, were input to the four LSTM models trained with modality-specific training data. Additionally, using the performance under single-modality data as benchmarks, respectively, the improvement rates of accuracy were calculated, further enabling quantification of performance improvement through multimodal fusion. The results of the above procedures are listed in Table 5.

According to the performance metrics of the LSTM model under multimodal data from the two validation experiments, the prediction accuracy under multimodal-fused data, with 94.9% under insufficient illumination and 91.1% under occlusion, were

**Table 5.** Performance metrics of the LSTM model for the examination of generalizability

| Environmental conditions  | Test data        | Performance metrics |            |
|---------------------------|------------------|---------------------|------------|
|                           |                  | Accuracy            | Boost rate |
| Insufficient illumination | Multimodal-fused | 0.949               | —          |
|                           | EEG + JP         | 0.653               | 45.3%      |
|                           | EEG + MT         | 0.892               | 6.4%       |
|                           | JP + MT          | 0.546               | 73.8%      |
|                           | EEG only         | 0.822               | 15.5%      |
|                           | JP only          | 0.305               | 211.2%     |
|                           | MT only          | 0.885               | 7.2%       |
| Occlusion                 | Multimodal-fused | 0.911               | —          |
|                           | EEG + JP         | 0.556               | 63.8%      |
|                           | EEG + MT         | 0.906               | 0.3%       |
|                           | JP + MT          | 0.659               | 38.2%      |
|                           | EEG only         | 0.862               | 5.7%       |
|                           | JP only          | 0.290               | 214.1%     |
|                           | MT only          | 0.901               | 1.1%       |

Note: JP = joint point data; EEG = EEG data; and MT = motion data.

lower than that of the training stage but significantly higher than those under single-modality data. In addition, the obtained boost rates indicated that the classification accuracy under multimodal-fused data was improved by 211.2%, 15.5%, and 7.2% under insufficient illumination and 214.1%, 5.7%, and 1.1% under occlusion compared with those under only joint point data, EEG data, and motion data. Therefore, models trained using the multimodal fusion were verified to be generalizable under adverse environmental conditions, achieving relatively high performance compared with those trained under single-modality data.

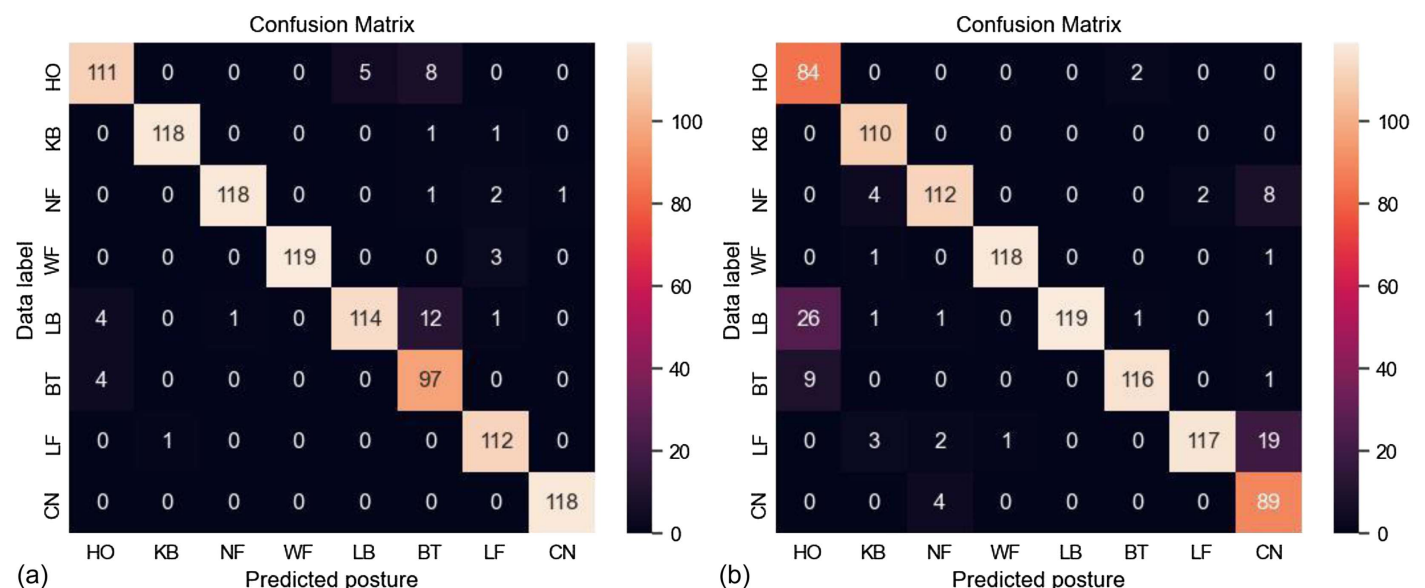
Moreover, regardless of insufficient illumination or occlusion, the prediction accuracies under joint point data were lower than those under other single-modality data. This objectively reflected the fact that in previous studies, computer vision-based recognition was significantly influenced by adverse environments (Yu et al. 2021; Huo and Hendriks 2012). Furthermore, confusion matrixes under multimodal-fused data were established and are shown in Fig. 8, for similarly identifying the postures confused with others. Under conditions of insufficient illumination, the LSTM model achieved optimal precision for all postures under the validation dataset, except for BT (body twisting on the ladder), which had 22 incorrectly predicted observations. Under the condition of occlusion, the prediction of the HO and CN (curled up in a narrow area) had relatively low precision, with 35 and 30 incorrectly predicted observations, respectively.

## Contributions and Limitations

### Contributions and Implications

Developing a multimodal-based approach to recognize and classify awkward working postures in construction, this study provides the theoretical and practical implications for both researchers and practitioners.

1. Theoretically, this study pioneers the utilization of advanced multimodal techniques in the construction domain. Not only does this study develop a referenceable framework based on multimodal fusion for classification, but also reveals intermodality incompatibility. Thus, this study contributes to ergonomic risk assessment in the



**Fig. 8.** Confusion matrixes under multimodal-fused data from adverse environmental conditions: (a) insufficient illumination; and (b) occlusion.

construction field. Specifically, using multimodal data collected by wearable sensors and RGB cameras, a feasible method for more accurate recognition of awkward working postures, is developed. This study aimed to address the limitations of computer vision-based posture identification in adverse environments considered, including insufficient illumination and occlusion. A potential improvement method based on multimodal technology is proposed, in contrast to other studies implemented only through optimizing algorithms. Additionally, the steps undertaken during the implementation of this study, which include data acquisition and processing, feature-level fusion, and comparative analysis of performance, can serve as a valuable reference for future studies. These steps can be utilized to validate and further improve upon the findings of this research.

2. Practically, this study successfully developed a multimodal-based approach for recognizing and classifying awkward working postures. The approach demonstrates impressive performance, achieving the highest classification accuracy of 99.6%. Even in adverse environments, it maintains a relatively high accuracy of 94.9%. This indicates that the approach not only exhibits superior performance but also possesses generalizability, enabling proactive control of WMSD risks in dynamic and complex construction sites. Furthermore, the implementation of a dedicated wearable sensor system combined with an RGB camera, which includes devices like Emotiv Epoc X, 12 resistive thin-film pressure sensors, and an IMU sensor, serves as a replicable and feasible solution for data acquisition. This system can be adopted in other projects aiming to develop and enhance management systems for ensuring the safety and health of construction workers. By effectively mitigating WMSD risks, this approach contributes to the overall improvement of worker safety and health in the construction industry.

### Limitations and Future Work

In the process of conducting this multimodal-based study, the following limitations emerged along with the development of the recognition and classification approach.

1. Limitations in the data acquisition: In this study, a workflow for data acquisition was developed, and relevant data were collected using wearable sensors and RGB cameras. However, there are opportunities for improvement in future studies, such as optimizing the experimental procedure, increasing the number of participants, and enhancing the experimental environment. Additionally, the adopted apparatus could be upgraded. For instance, the pressure sensors could be replaced with a more integrated OpenGo system, which is commonly used in relevant studies (Antwi-Afari et al. 2022). This upgrade would enhance the accuracy and reliability of the data collected during posture assessment.
2. Limitations in the multimodal fusion strategy: Early fusion, specifically feature-level fusion, is used as the multimodal fusion strategy in this study. This approach enables early correlation between different multimodal features, resulting in outstanding performance. However, it is worth noting that there is a lack of performance metric support for the selection of the early fusion strategy. Therefore, a comparative analysis of models trained using other fusion strategies is necessary to determine the optimal configuration.
3. Limitations in the selected networks: To classify various types of awkward working postures, three commonly used deep learning-based networks were trained in this study: MLP, RNN, and LSTM models. Among them, the LSTM model achieved the best performance and demonstrated good generalizability based on the output metrics. However, it is worth

mentioning that during prediction and validation, a few incorrect predictions or rejected observations still occurred. For future studies, it is recommended to explore and develop additional networks, such as Bi-LSTM or other variations, that have been successfully employed in existing studies. These networks can be further developed and tested to enhance the accuracy and reliability of posture classification.

### Conclusions

This study contributes to the existing body of knowledge on enhancing safety and health performance within the construction industry. To that end, the study introduced an innovative and automated approach that employs multimodal fusion technology to recognize and classify awkward working postures. This approach directly helps mitigate potential work-related musculoskeletal disorder (WMSD) risk factors during operations, effectively addressing limitations in previous research efforts. Specifically, this study conducted indoor experiments to acquire multimodal data, focusing on eight working postures identified through site investigation and literature review. A dedicated wearable sensor system consisting of an Emotiv Epoc X wearable EEG device, foot pressure sensors, and IMU sensors was employed. Following data preprocessing and segmentation, three deep-learning-based models (MLP, RNN, and LSTM) were trained using a feature-level multimodal fusion strategy. According to the experimental results and output metrics, the LSTM model achieved optimal prediction performance, attaining a remarkable accuracy of 99.6% and an F1-score of 99.7%. Comparing metrics obtained from single-modality and multimodal-fused training data revealed that the multimodal fusion approach employed in this study significantly improved model performance. To examine the generalizability of the LSTM model, test data collected from two sets of experiments conducted under adverse environmental conditions were used. The results indicated that model performance declined in adverse environmental conditions, particularly in accuracy, when using only joint point data. This is consistent with the fact that computer vision-based recognition is significantly affected by environmental changes. However, the multimodal fusion strategy enhanced the model's generalizability, resulting in improved prediction performance of 94.9% accuracy in insufficient illumination and 91.1% accuracy in occlusion scenarios. Despite the excellent performance and generalizability demonstrated by the developed multimodal technology-based approach, there are limitations in data acquisition, multimodal fusion processes, and model training. It is crucial to address these limitations in future research. Importantly, the methodology proposed in this study is replicable, allowing for further refinement and enhancement of prediction performance using multimodal technology.

### Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

### Acknowledgments

This research was funded by the National Natural Science Foundation of China (Nos. 72101054, 51978164, and 72271122) and Humanities and Social Sciences Youth Foundation, Ministry of Education of the People's Republic of China (20YJCZH182). The authors extend appreciation to Prof. Pingbo Tang from

## References

- Abobakr, A., D. Nahavandi, M. Hossny, J. Iskander, M. Attia, S. Nahavandi, and M. Smets. 2019. "RGB-D ergonomic assessment system of adopted working postures." *Appl. Ergon.* 80 (Oct): 75–88. <https://doi.org/10.1016/j.apergo.2019.05.004>.
- Angah, O., and A. Y. Chen. 2020. "Tracking multiple construction workers through deep learning and the gradient based method with re-matching based on multi-object tracking accuracy." *Autom. Constr.* 119 (Nov): 103308. <https://doi.org/10.1016/j.autcon.2020.103308>.
- Antwi-Afari, M. F., and H. Li. 2018. "Fall risk assessment of construction workers based on biomechanical gait stability parameters using wearable insole pressure system." *Adv. Eng. Inf.* 38 (Oct): 683–694. <https://doi.org/10.1016/j.aei.2018.10.002>.
- Antwi-Afari, M. F., H. Li, D. J. Edwards, E. A. Pärn, J. Seo, and A. Y. L. Wong. 2017. "Biomechanical analysis of risk factors for work-related musculoskeletal disorders during repetitive lifting task in construction workers." *Autom. Constr.* 83 (Nov): 41–47. <https://doi.org/10.1016/j.autcon.2017.07.007>.
- Antwi-Afari, M. F., H. Li, J. Seo, and A. Y. L. Wong. 2018a. "Automated detection and classification of construction workers' loss of balance events using wearable insole pressure sensors." *Autom. Constr.* 96 (Dec): 189–199. <https://doi.org/10.1016/j.autcon.2018.09.010>.
- Antwi-Afari, M. F., H. Li, Y. Yu, and L. Kong. 2018b. "Wearable insole pressure system for automated detection and classification of awkward working postures in construction workers." *Autom. Constr.* 96 (Dec): 433–441. <https://doi.org/10.1016/j.autcon.2018.10.004>.
- Antwi-Afari, M. F., Y. Qarout, R. Herzallah, S. Anwer, W. Umer, Y. Zhang, and P. Manu. 2022. "Deep learning-based networks for automated recognition and classification of awkward working postures in construction using wearable insole sensor data." *Autom. Constr.* 136 (Apr): 104181. <https://doi.org/10.1016/j.autcon.2022.104181>.
- Baduge, S. K., S. Thilakarathna, J. S. Perera, M. Arashpour, P. Sharafi, B. Teodosio, A. Shringi, and P. Mendis. 2022. "Artificial intelligence and smart vision for building and construction 4.0: Machine and deep learning methods and applications." *Autom. Constr.* 141 (Sep): 104440. <https://doi.org/10.1016/j.autcon.2022.104440>.
- Bingöl, Ö., and M. Ekinici. 2017. "Stereo-based palmprint recognition in various 3D postures." *Expert Syst. Appl.* 78 (Jul): 74–88. <https://doi.org/10.1016/j.eswa.2017.01.025>.
- Boeschoten, S., C. Catal, B. Tekinerdogan, A. Lommen, and M. Blokland. 2023. "The automation of the development of classification models and improvement of model quality using feature engineering techniques." *Expert Syst. Appl.* 213 (Mar): 118912. <https://doi.org/10.1016/j.eswa.2022.118912>.
- Brandl, C., A. Mertens, and C. M. Schlick. 2017. "Effect of sampling interval on the reliability of ergonomic analysis using the Ovako working posture analysing system (OWAS)." *Int. J. Ind. Ergon.* 57 (Jan): 68–73. <https://doi.org/10.1016/j.ergon.2016.11.013>.
- Buchholz, B., V. Paquet, L. Punnett, D. Lee, and S. Moir. 1996. "PATH: A work sampling-based approach to ergonomic job analysis for construction and other non-repetitive work." *Appl. Ergon.* 27 (3): 177–187. [https://doi.org/10.1016/0003-6870\(95\)00078-X](https://doi.org/10.1016/0003-6870(95)00078-X).
- Chen, H., X. Luo, Z. Zheng, and J. Ke. 2019. "A proactive workers' safety risk evaluation framework based on position and posture data fusion." *Autom. Constr.* 98 (Feb): 275–288. <https://doi.org/10.1016/j.autcon.2018.11.026>.
- Chen, J., J. Qiu, and C. Ahn. 2017. "Construction worker's awkward posture recognition through supervised motion tensor decomposition." *Autom. Constr.* 77 (May): 67–81. <https://doi.org/10.1016/j.autcon.2017.01.020>.
- Cutlip, R., H. Hsiao, R. Garcia, E. Becker, and B. Mayeux. 2000. "A comparison of different postures for scaffold end-frame disassembly." *Appl. Ergon.* 31 (5): 507–513. [https://doi.org/10.1016/S0003-6870\(00\)00016-8](https://doi.org/10.1016/S0003-6870(00)00016-8).
- Dahlberg, R., L. Karlqvist, C. Bildt, and K. Nykvist. 2004. "Do work technique and musculoskeletal symptoms differ between men and women performing the same type of work tasks?" *Appl. Ergon.* 35 (6): 521–529. <https://doi.org/10.1016/j.apergo.2004.06.008>.
- David, G. C. 2005. "Ergonomic methods for assessing exposure to risk factors for work-related musculoskeletal disorders." *Occup. Med.* 55 (3): 190–199. <https://doi.org/10.1093/occmed/kqi082>.
- de Vries, A. W., F. Krause, and M. P. de Looze. 2021. "The effectivity of a passive arm support exoskeleton in reducing muscle activation and perceived exertion during plastering activities." *Ergonomics* 64 (6): 712–721. <https://doi.org/10.1080/00140139.2020.1868581>.
- Dockrell, S., E. O'Grady, K. Bennett, C. Mullarkey, R. McConnell, R. Ruddy, S. Twomey, and C. Flannery. 2012. "An investigation of the reliability of rapid upper limb assessment (RULA) as a method of assessment of children's computing posture." *Appl. Ergon.* 43 (3): 632–636. <https://doi.org/10.1016/j.apergo.2011.09.009>.
- Fang, W., L. Ding, H. Luo, and P. E. D. Love. 2018. "Falls from heights: A computer vision-based approach for safety harness detection." *Autom. Constr.* 91 (Jul): 53–61. <https://doi.org/10.1016/j.autcon.2018.02.018>.
- Fang, W., B. Zhong, N. Zhao, P. E. D. Love, H. Luo, J. Xue, and S. Xu. 2019. "A deep learning-based approach for mitigating falls from height with computer vision: Convolutional neural network." *Adv. Eng. Inf.* 39 (Jan): 170–177. <https://doi.org/10.1016/j.aei.2018.12.005>.
- Gandhi, A., K. Adhvaryu, S. Poria, E. Cambria, and A. Hussain. 2023. "Multimodal sentiment analysis: A systematic review of history, datasets, multimodal fusion methods, applications, challenges and future directions." *Inf. Fusion* 91 (Mar): 424–444. <https://doi.org/10.1016/j.inffus.2022.09.025>.
- Guo, H., Y. Yu, and M. Skitmore. 2017. "Visualization technology-based construction safety management: A review." *Autom. Constr.* 73 (Jan): 135–144. <https://doi.org/10.1016/j.autcon.2016.10.004>.
- Hignett, S., and L. McAtamney. 2000. "Rapid entire body assessment (REBA)." *Appl. Ergon.* 31 (2): 201–205. [https://doi.org/10.1016/S0003-6870\(99\)00039-3](https://doi.org/10.1016/S0003-6870(99)00039-3).
- Huo, F., and E. A. Hendriks. 2012. "Multiple people tracking and pose estimation with occlusion estimation." *Comput. Vis. Image Understanding* 116 (5): 634–647. <https://doi.org/10.1016/j.cviu.2011.12.006>.
- Jebelli, H., S. Hwang, and S. Lee. 2018. "EEG-based workers' stress recognition at construction sites." *Autom. Constr.* 93 (Sep): 315–324. <https://doi.org/10.1016/j.autcon.2018.05.027>.
- Jeelani, I., K. Asadi, H. Ramshankar, K. Han, and A. Albert. 2021. "Real-time vision-based worker localization & hazard detection for construction." *Autom. Constr.* 121 (Jan): 103448. <https://doi.org/10.1016/j.autcon.2020.103448>.
- Jeon, J., and H. Cai. 2021. "Classification of construction hazard-related perceptions using wearable electroencephalogram and virtual reality." *Autom. Constr.* 132 (Dec): 103975. <https://doi.org/10.1016/j.autcon.2021.103975>.
- Jeon, J., and H. Cai. 2022. "Multi-class classification of construction hazards via cognitive states assessment using wearable EEG." *Adv. Eng. Inf.* 53 (Aug): 101646. <https://doi.org/10.1016/j.aei.2022.101646>.
- Jung, J.-Y., H.-Y. Cho, and C.-K. Kang. 2020. "Brain activity during a working memory task in different postures: An EEG study." *Ergonomics* 63 (11): 1359–1370. <https://doi.org/10.1080/00140139.2020.1784467>.
- Ke, J., M. Zhang, X. Luo, and J. Chen. 2021. "Monitoring distraction of construction workers caused by noise using a wearable electroencephalography (EEG) device." *Autom. Constr.* 125 (May): 103598. <https://doi.org/10.1016/j.autcon.2021.103598>.
- Kee, D. 2021. "Comparison of OWAS, RULA and REBA for assessing potential work-related musculoskeletal disorders." *Int. J. Ind. Ergon.* 83 (May): 103140. <https://doi.org/10.1016/j.ergon.2021.103140>.
- Kittusamy, N. K., and B. Buchholz. 2004. "Whole-body vibration and postural stress among operators of construction equipment: A literature review." *J. Saf. Res.* 35 (3): 255–261. <https://doi.org/10.1016/j.jsr.2004.03.014>.
- Kong, L., H. Li, Y. Yu, H. Luo, M. Skitmore, and M. F. Antwi-Afari. 2018. "Quantifying the physical intensity of construction workers, a mechanical energy approach." *Adv. Eng. Inf.* 38 (Oct): 404–419. <https://doi.org/10.1016/j.aei.2018.08.005>.

- Kumar, P., A. K. Das, and S. Halder. 2020. "Time-domain HRV analysis of ECG signal under different body postures." *Procedia Comput. Sci.* 167 (Jan): 1705–1710. <https://doi.org/10.1016/j.procs.2020.03.435>.
- Majumder, N., D. Hazarika, A. Gelbukh, E. Cambria, and S. Poria. 2018. "Multimodal sentiment analysis using hierarchical fusion with context modeling." *Knowledge-Based Syst.* 161 (Dec): 124–133. <https://doi.org/10.1016/j.knosys.2018.07.041>.
- Man, S. S., A. H. Chan, and H. M. Wong. 2017. "Risk-taking behaviors of Hong Kong construction workers—A thematic study." *Saf. Sci.* 98 (Oct): 25–36. <https://doi.org/10.1016/j.ssci.2017.05.004>.
- Massiris-Fernández, M., J. Á. Fernández, J. M. Bajo, and C. A. Delrieux. 2020. "Ergonomic risk assessment based on computer vision and machine learning." *Comput. Ind. Eng.* 149 (Nov): 106816. <https://doi.org/10.1016/j.cie.2020.106816>.
- Mora, D. C., C. M. Miles, H. Chen, S. A. Quandt, P. Summers, and T. A. Arcury. 2016. "Prevalence of musculoskeletal disorders among immigrant Latino farmworkers and non-farmworkers in North Carolina." *Arch. Environ. Occup. Health* 71 (3): 136–143. <https://doi.org/10.1080/19338244.2014.988676>.
- Nunes, I. L., and P. M. Bush. 2012. "Work-related musculoskeletal disorders assessment and prevention." In Vol. 1 of *Ergonomics: A systems approach*, 30. Rijeka, Croatia: InTech.
- Pan, Y., and L. Zhang. 2021. "Roles of artificial intelligence in construction engineering and management: A critical review and future trends." *Autom. Constr.* 122 (Feb): 103517. <https://doi.org/10.1016/j.autcon.2020.103517>.
- Poria, S., H. Peng, A. Hussain, N. Howard, and E. Cambria. 2017. "Ensemble application of convolutional neural networks and multiple kernel learning for multimodal sentiment analysis." *Neurocomputing* 261 (Oct): 217–230. <https://doi.org/10.1016/j.neucom.2016.09.117>.
- Qi, H., Z. Zhou, N. Li, and C. Zhang. 2022. "Construction safety performance evaluation based on data envelopment analysis (DEA) from a hybrid perspective of cross-sectional and longitudinal." *Saf. Sci.* 146 (Feb): 105532. <https://doi.org/10.1016/j.ssci.2021.105532>.
- Roberts, D., W. T. Calderon, S. Tang, and M. Golparvar-Fard. 2020. "Vision-Based construction worker activity analysis informed by body posture." *J. Comput. Civ. Eng.* 34 (4): 04020017. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000898](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000898).
- Roy, R., D. Sikdar, and M. Mahadevappa. 2020. "Chaotic behaviour of EEG responses with an identical grasp posture." *Comput. Biol. Med.* 123 (Aug): 103822. <https://doi.org/10.1016/j.combiomed.2020.103822>.
- Ryu, J., M. M. Diraneyya, C. T. Haas, and E. Abdel-Rahman. 2021. "Analysis of the limits of automated rule-based ergonomic assessment in bricklaying." *J. Constr. Eng. Manage.* 147 (2): 04020163. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001978](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001978).
- Ryu, J., T. McFarland, C. T. Haas, and E. Abdel-Rahman. 2022. "Automatic clustering of proper working postures for phases of movement." *Autom. Constr.* 138 (Jun): 104223. <https://doi.org/10.1016/j.autcon.2022.104223>.
- Sarkar, A., A. Singh, and R. Chakraborty. 2022. "A deep learning-based comparative study to track mental depression from EEG data." *Neurosci. Inf.* 2 (4): 100039. <https://doi.org/10.1016/j.neuri.2022.100039>.
- Schall, M. C., Jr., N. B. Fethke, H. Chen, S. Oyama, and D. I. Doupbrate. 2016. "Accuracy and repeatability of an inertial measurement unit system for field-based occupational studies." *Ergonomics* 59 (4): 591–602. <https://doi.org/10.1080/00140139.2015.1079335>.
- Smith, L. N. 2017. "Cyclical learning rates for training neural networks." In *Proc., 2017 IEEE Winter Conf. on Applications of Computer Vision (WACV)*, 464–472. New York: IEEE.
- Sobeih, T., O. Salem, A. Genaïdy, T. Abdelhamid, and R. Shell. 2009. "Psychosocial factors and musculoskeletal disorders in the construction industry." *J. Constr. Eng. Manage.* 135 (4): 267–277. [https://doi.org/10.1061/\(ASCE\)0733-9364\(2009\)135:4\(267\)](https://doi.org/10.1061/(ASCE)0733-9364(2009)135:4(267)).
- Son, H., and C. Kim. 2021. "Integrated worker detection and tracking for the safe operation of construction machinery." *Autom. Constr.* 126 (Jun): 103670. <https://doi.org/10.1016/j.autcon.2021.103670>.
- Spielholz, P., B. Silverstein, M. Morgan, H. Checkoway, and J. Kaufman. 2001. "Comparison of self-report, video observation and direct measurement methods for upper extremity musculoskeletal disorder physical risk factors." *Ergonomics* 44 (6): 588–613. <https://doi.org/10.1080/00140130118050>.
- Spielholz, P., B. Silverstein, and M. Stuart. 1999. "Reproducibility of a self-report questionnaire for upper extremity musculoskeletal disorder risk factors." *Appl. Ergon.* 30 (5): 429–433. [https://doi.org/10.1016/S0003-6870\(98\)00049-0](https://doi.org/10.1016/S0003-6870(98)00049-0).
- Thakur, D., and S. Biswas. 2022. "An integration of feature extraction and guided regularized random forest feature selection for smartphone based human activity recognition." *J. Network Comput. Appl.* 204 (Aug): 103417. <https://doi.org/10.1016/j.jnca.2022.103417>.
- Toyoshima, K., T. Yasunaga, C. Yukawa, Y. Naga, N. Saito, T. Oda, and L. Barolli. 2023. "Analysis of a soldering motion for dozing state and attention posture detection." In *Proc., Advances on P2P, Parallel, Grid, Cloud and Internet Computing*, 146–153. Berlin: Springer. [https://doi.org/10.1007/978-3-031-19945-5\\_14](https://doi.org/10.1007/978-3-031-19945-5_14).
- Umer, W., M. F. Antwi-Afari, H. Li, G. P. Y. Szeto, and A. Y. L. Wong. 2018. "The prevalence of musculoskeletal symptoms in the construction industry: A systematic review and meta-analysis." *Int. Arch. Occup. Environ. Health* 91 (Feb): 125–144. <https://doi.org/10.1007/s00420-017-1273-4>.
- Umer, W., H. Li, G. P. Y. Szeto, and A. Y. L. Wong. 2017. "Identification of biomechanical risk factors for the development of lower-back disorders during manual rebar tying." *J. Comput. Civ. Eng.* 143 (1): 04016080. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001208](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001208).
- Valero, E., A. Sivanathan, F. Bosché, and M. Abdel-Wahab. 2017. "Analysis of construction trade worker body motions using a wearable and wireless motion sensor network." *Autom. Constr.* 83 (Nov): 48–55. <https://doi.org/10.1016/j.autcon.2017.08.001>.
- Vieira, E. R., and S. Kumar. 2004. "Working postures: A literature review." *J. Occup. Rehabil.* 14 (Jun): 143–159. <https://doi.org/10.1023/B:JOOR.0000018330.46029.05>.
- Wang, J., D. Chen, M. Zhu, and Y. Sun. 2021. "Risk assessment for musculoskeletal disorders based on the characteristics of work posture." *Autom. Constr.* 131 (Nov): 103921. <https://doi.org/10.1016/j.autcon.2021.103921>.
- Wu, J., N. Cai, W. Chen, H. Wang, and G. Wang. 2019. "Automatic detection of hardhats worn by construction personnel: A deep learning approach and benchmark dataset." *Autom. Constr.* 106 (Oct): 102894. <https://doi.org/10.1016/j.autcon.2019.102894>.
- Xiao, F., Z. Zhang, C. Liu, and Y. Wang. 2023. "Human motion intention recognition method with visual, audio, and surface electromyography modalities for a mechanical hand in different environments." *Biomed. Signal Process. Control* 79 (Jan): 104089. <https://doi.org/10.1016/j.bspc.2022.104089>.
- Xing, X., B. Zhong, H. Luo, T. Rose, J. Li, and M. F. Antwi-Afari. 2020. "Effects of physical fatigue on the induction of mental fatigue of construction workers: A pilot study based on a neurophysiological approach." *Autom. Constr.* 120 (Dec): 103381. <https://doi.org/10.1016/j.autcon.2020.103381>.
- Yan, X., H. Li, A. R. Li, and H. Zhang. 2017a. "Wearable IMU-based real-time motion warning system for construction workers' musculoskeletal disorders prevention." *Autom. Constr.* 74 (Feb): 2–11. <https://doi.org/10.1016/j.autcon.2016.11.007>.
- Yan, X., H. Li, C. Wang, J. Seo, H. Zhang, and H. Wang. 2017b. "Development of ergonomic posture recognition technique based on 2D ordinary camera for construction hazard prevention through view-invariant features in 2D skeleton motion." *Adv. Eng. Inf.* 34 (Oct): 152–163. <https://doi.org/10.1016/j.aei.2017.11.001>.
- Yan, X., H. Li, H. Zhang, and T. M. Rose. 2018. "Personalized method for self-management of trunk postural ergonomic hazards in construction rebar ironwork." *Adv. Eng. Inf.* 37 (Aug): 31–41. <https://doi.org/10.1016/j.aei.2018.04.013>.
- Yang, K., C. R. Ahn, and H. Kim. 2020. "Deep learning-based classification of work-related physical load levels in construction." *Adv. Eng. Inf.* 45 (Aug): 101104. <https://doi.org/10.1016/j.aei.2020.101104>.
- Yang, K., C. R. Ahn, M. C. Vuran, and H. Kim. 2017. "Collective sensing of workers' gait patterns to identify fall hazards in construction." *Autom. Constr.* 82 (Oct): 166–178. <https://doi.org/10.1016/j.autcon.2017.04.010>.

- Yang, M., C. Wu, Y. Guo, R. Jiang, F. Zhou, J. Zhang, and Z. Yang. 2023. "Transformer-based deep learning model and video dataset for unsafe action identification in construction projects." *Autom. Constr.* 146 (Feb): 104703. <https://doi.org/10.1016/j.autcon.2022.104703>.
- Ying, W., W. Shou, J. Wang, W. Shi, Y. Sun, D. Ji, H. Gai, X. Wang, and M. Chen. 2021. "Automatic scaffolding workplace assessment for activity analysis through machine learning." *Appl. Sci.* 11 (9): 4143. <https://doi.org/10.3390/app11094143>.
- Yu, Y., H. Guo, Q. Ding, H. Li, and M. Skitmore. 2017. "An experimental study of real-time identification of construction workers' unsafe behaviors." *Autom. Constr.* 82 (Oct): 193–206. <https://doi.org/10.1016/j.autcon.2017.05.002>.
- Yu, Y., W. Umer, X. Yang, and M. F. Antwi-Afari. 2021. "Posture-related data collection methods for construction workers: A review." *Autom. Constr.* 124 (Apr): 103538. <https://doi.org/10.1016/j.autcon.2020.103538>.
- ZakerJafari, H. R., and M. H. YektaKooshali. 2018. "Work-related musculoskeletal disorders in Iranian dentists: A systematic review and meta-analysis." *Saf. Health Work* 9 (1): 1–9. <https://doi.org/10.1016/j.shaw.2017.06.006>.
- Zhang, H., X. Yan, and H. Li. 2018. "Ergonomic posture recognition using 3D view-invariant features from single ordinary camera." *Autom. Constr.* 94 (Oct): 1–10. <https://doi.org/10.1016/j.autcon.2018.05.033>.
- Zhang, W., J. Yu, H. Hu, H. Hu, and Z. Qin. 2020. "Multimodal feature fusion by relational reasoning and attention for visual question answering." *Inf. Fusion* 55 (Mar): 116–126. <https://doi.org/10.1016/j.inffus.2019.08.009>.
- Zhang, X., J. Fan, T. Peng, P. Zheng, C. K. M. Lee, and R. Tang. 2022. "A privacy-preserving and unobtrusive sitting posture recognition system via pressure array sensor and infrared array sensor for office workers." *Adv. Eng. Inf.* 53 (Aug): 101690. <https://doi.org/10.1016/j.aei.2022.101690>.
- Zhao, J., and E. Obonyo. 2020. "Convolutional long short-term memory model for recognizing construction workers' postures from wearable inertial measurement units." *Adv. Eng. Inf.* 46 (Oct): 101177. <https://doi.org/10.1016/j.aei.2020.101177>.
- Zhao, J., and E. Obonyo. 2021. "Applying incremental deep neural networks-based posture recognition model for ergonomics risk assessment in construction." *Adv. Eng. Inf.* 50 (Oct): 101374. <https://doi.org/10.1016/j.aei.2021.101374>.